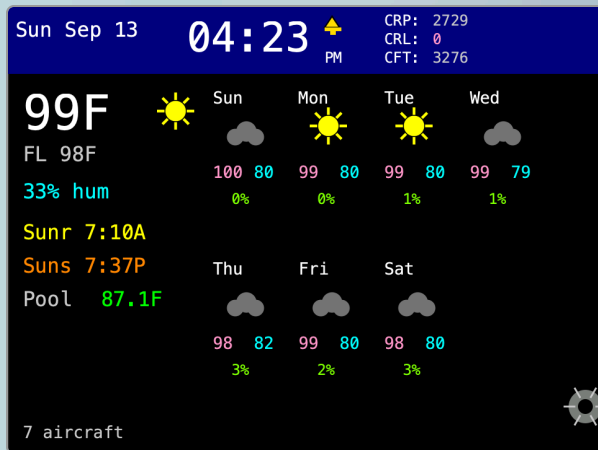


## What it shows

- **Clock & date** — always on top.
- **Weather** — current temperature, feels-like, humidity, sunrise/sunset, a 7-day forecast, and temperature history graphs (Day / Week / Month / Year).
- **Flights overhead** (optional — on by default) — a mini radar of live aircraft near you: callsign, route, altitude, speed, distance, airline logo, and the plane's actual ground track.
- **Pool temperature** (optional — off by default) — current water temperature and history graphs, if you have a Govee pool thermometer.
- **Sleep mode** — the screen sleeps overnight and wakes on touch.
- **Alarms** — up to 6 weekday-scheduled alarms with an LED + sound alert.
- **Self-updates** — new firmware installs itself over WiFi.



## Resetting (Settings → Reset)

Every option asks you to confirm and says exactly what it clears:

- **Restart** — reboots only; clears nothing.
- **Graph Data** — clears pool and weather temperature history.
- **Settings** — clears settings and credentials (keeps touch calibration).
- **Factory Reset** — clears *everything*: settings, credentials, files, history, and touch calibration. The next boot recalibrates touch and asks for WiFi again.
- **Cancel** — backs out without changing anything.

⚠ **Factory Reset erases any pre-loaded airline logos** — restoring logos requires regenerating and reloading them from a development computer.

## Troubleshooting

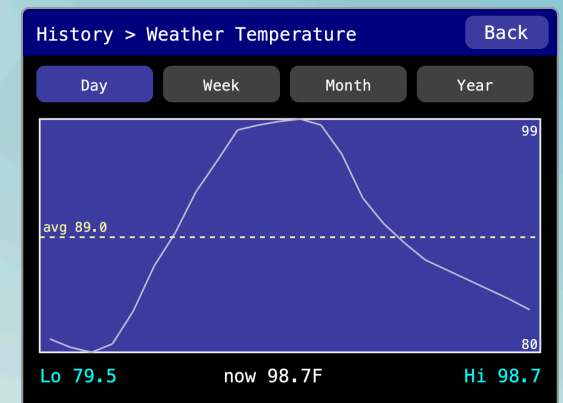
- **Taps land in the wrong spot** — press and hold anywhere on the screen for **10 seconds** to recalibrate.
- **Wrong weather or no flights nearby** — fix your location under **Settings → Location**.
- **New WiFi network** — re-enter it under **Settings → Network**.



# cyd-dashboard

Weather · Flights · Pool Temp · Alarms

A small self-contained touchscreen dashboard for **Cheap Yellow Display (CYD)** boards — the 2.8" 2432S028R or the 4" E32R40T. Once it's set up it runs on your WiFi — no computer needed.



Firmware v1.18.0+

[github.com/improving-minnesota/cyd-dashboard](https://github.com/improving-minnesota/cyd-dashboard)

## Getting started

1. **Plug it in.** On a brand-new device (or after a Factory Reset) it first asks you to tap a few crosshairs to calibrate the touchscreen — this is skipped if calibration is already saved.
2. **Enter WiFi.** Type your network name and password on the on-screen keyboard. (Change it later under **Settings** → **Network**.)
3. Done — the dashboard appears and starts loading weather and flights.

## Set your location

Weather and flights need to know where you are. On first boot the device guesses your location from your internet connection — check that it's right:

Open **Settings** → **Location**, then use whichever is easiest:

- **Search Address** — type your city or address and pick the match.
- **Set** — type your latitude and longitude directly.
- **Find by IP** — guess again from your connection.

## OpenSky API key (recommended)

Flights work out of the box, but a free OpenSky account raises your daily limit and removes the yellow "anonymous" warning.

1. Create a free account at [opensky-network.org](https://opensky-network.org).
2. Go to **My OpenSky** → **Account** and create an **API client** — that downloads a file containing your **client ID** and **client secret**.
3. On the device: open **Settings** → **Flight Tracker**, page right to the **OpenSky Credentials** screen, and enter both.

The three small readouts in the header — **CRP**, **CRL**, **CFT** — show your remaining daily OpenSky credits (tap them for detail). Grey is healthy, yellow is low, pink is nearly out. Credits reset daily.

## Govee pool thermometer (optional)

Only if you own a Govee pool thermometer:

1. Create a free developer account at [developer.govee.com](https://developer.govee.com) and generate an **API key**.
2. On the device: open **Settings** → **Pool Temp**, turn it **on**, enter the key, tap **Fetch Devices**, and pick your thermometer.

## Using the dashboard

The main screen isn't just a display — several spots are **buttons you can tap**:

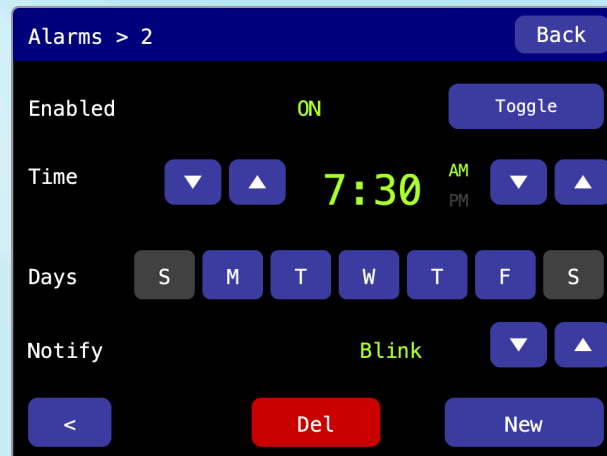
- **Cog (bottom-right)** — open settings.
- **Weather temperature (top-left)** — weather history graph.
- **Pool reading** — pool temperature history graph.
- **Aircraft count (bottom-left)** — details of the last overhead flight.
- **CRP / CRL / CFT (header)** — OpenSky credits detail.
- **Clock/date (header)** — open the alarm editor.
- Any settings or graph screen returns to the dashboard after 2 minutes untouched.

## Alarms

Up to 6, each with its own weekdays and LED pattern (Blink, Rapid, Double, Colors, Pulse). Set the time with the ▼ / ▲ arrow buttons — hour on the left, minute on the right (the hour rolls through AM/PM) — or tap the time to type it. A bell icon sits beside the clock while any alarm is on. **Del** asks for confirmation before removing an alarm.

When one fires: **Dismiss** stops it for today; **Snooze** refires in 5 min (up to 3 snoozes, then it dismisses for the day). Alarms fire on time even during Sleep Mode — the device wakes itself at the alarm time — and a snoozed or missed alarm still fires after a power loss.

 Sound requires an **external speaker** plugged into the board's small JST port — the display has no built-in speaker, so without one alarms flash the LED only.



## Settings at a glance

Defaults shown in parentheses.

- **General** — theme color picked from swatch rows (tap a hue, then a shade or grey — colors the header and screen buttons); auto-update (on); units (Imperial ft/mph/mi, Metric m/kts/km, or Aviation ft/kts/nm — also sets weather and pool temps to °F or °C); clock (12-hour with AM/PM, or 24-hour).
- **Location** — your coordinates (auto-guessed on first boot).
- **Network** — WiFi network (shows *Scanning* while it searches); DHCP or static IP; device hostname.
- **Flight Tracker** — enabled (on); radar radius (3.5, in mi/km/nm — how far away to look); altitude ceiling (15,000, in ft/m — planes above it are ignored); poll interval (30 s); timer bar (off); home airport (none); watched callsign (none); LED blink (on); Show IATA (on); OpenSky credentials.
- **Sleep Mode** — enabled (on); sleeps 10:00 PM – 8:00 AM; a touch wakes it for 5 minutes.
- **Pool Temp** — (off) enable, enter the API key, pick your thermometer.
- **Calibrate Touch** — rerun touch calibration.
- **About** — version, author, and firmware update status / install.
- **Reset** — see the back cover.

## LED & border colors

- **Red border / LED** — something needs attention: no WiFi, bad OpenSky credentials, exhausted OpenSky credits, or missing weather/pool data.
- **Yellow** — OpenSky is running anonymously; flights still work.
- **Flight blinks** — blue for any overhead flight; red when it's departing your home airport; green when arriving; yellow when both ends are your home airport; white for your watched callsign.

## Updates

The device updates itself over WiFi. **Auto-Update** (on by default) checks once a day; or open **Settings** → **About** and tap **Install** when a newer version is offered. If an update fails to start, it rolls back automatically.